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```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_and_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-selectedcolumns.csv")

print(df.head())
print(df.shape)
```

```
   Age Attrition     BusinessTravel  DailyRate              Department
    41       Yes      Travel_Rarely       1102                   Sales

\
0

1   49        No  Travel_Frequently        279  Research & Development

2   37       Yes      Travel_Rarely       1373  Research & Development

3   33        No  Travel_Frequently       1392  Research & Development

4   27        No      Travel_Rarely        591  Research & Development


   DistanceFromHome  Education EducationField  EmployeeCount
EmployeeNumbe
                  1          2  Life Sciences              1 0
                                                             1
                  8          1  Life Sciences              1 1
                                                             2
                  2          2          Other              1 2
                                                             4
                  3          4  Life Sciences              1 3
                                                             5
                  2          1        Medical              1 4
                                                             7

(1470, 10)
```

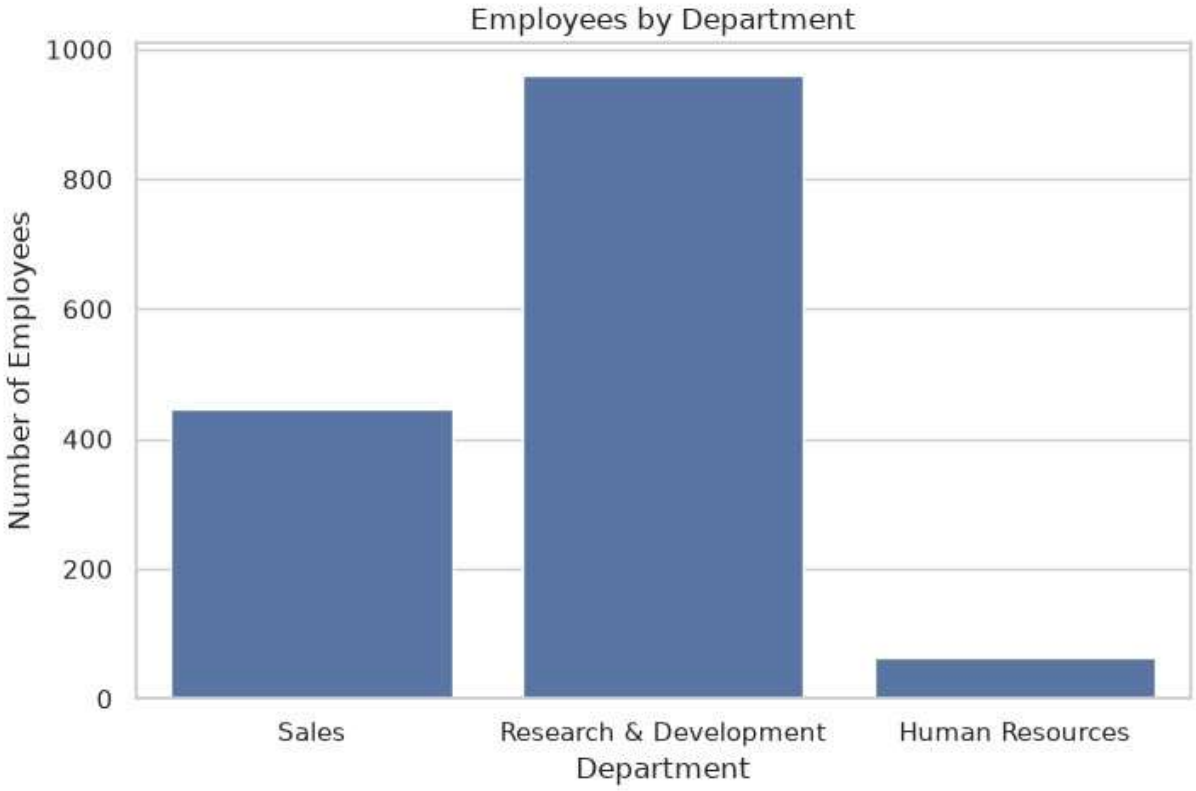
```
print(df.columns.tolist())
```

```
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
```

```
'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
'EmployeeNumber']
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df, x="Department")
```

```
plt.title("Employees by Department")
plt.xlabel("Department") plt.ylabel("Number of
Employees") plt.show()
```



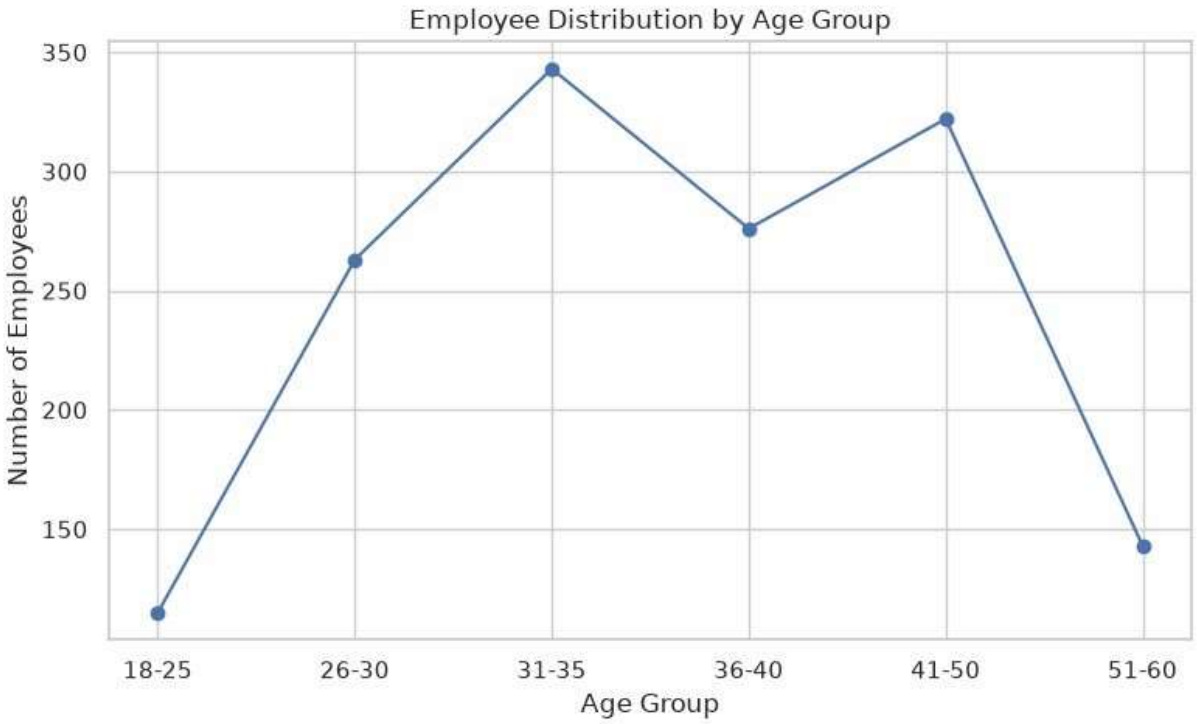

```
df["AgeGroup"] = pd.cut(
df["Age"],
    bins=[18, 25, 30, 35, 40, 50, 60],
    labels=["18-25", "26-30", "31-35", "36-40", "41-50", "51-60"]
)

age_count = df["AgeGroup"].value_counts().sort_index()

plt.figure(figsize=(9, 5))

plt.plot(age_count.index, age_count.values, marker="o")

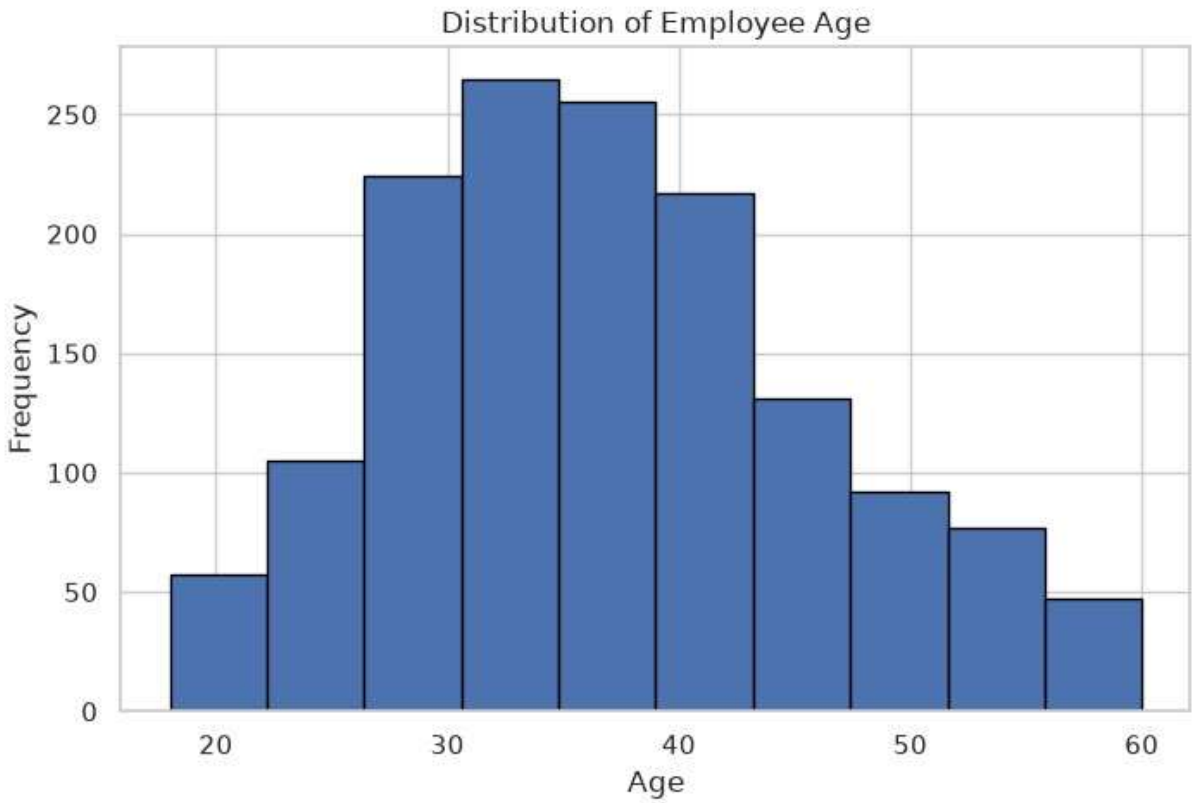
plt.title("Employee Distribution by Age Group")
plt.xlabel("Age Group") plt.ylabel("Number of
Employees")
plt.show()
```




```
plt.figure(figsize=(8, 5))

plt.hist(df["Age"], bins=10, edgecolor="black")

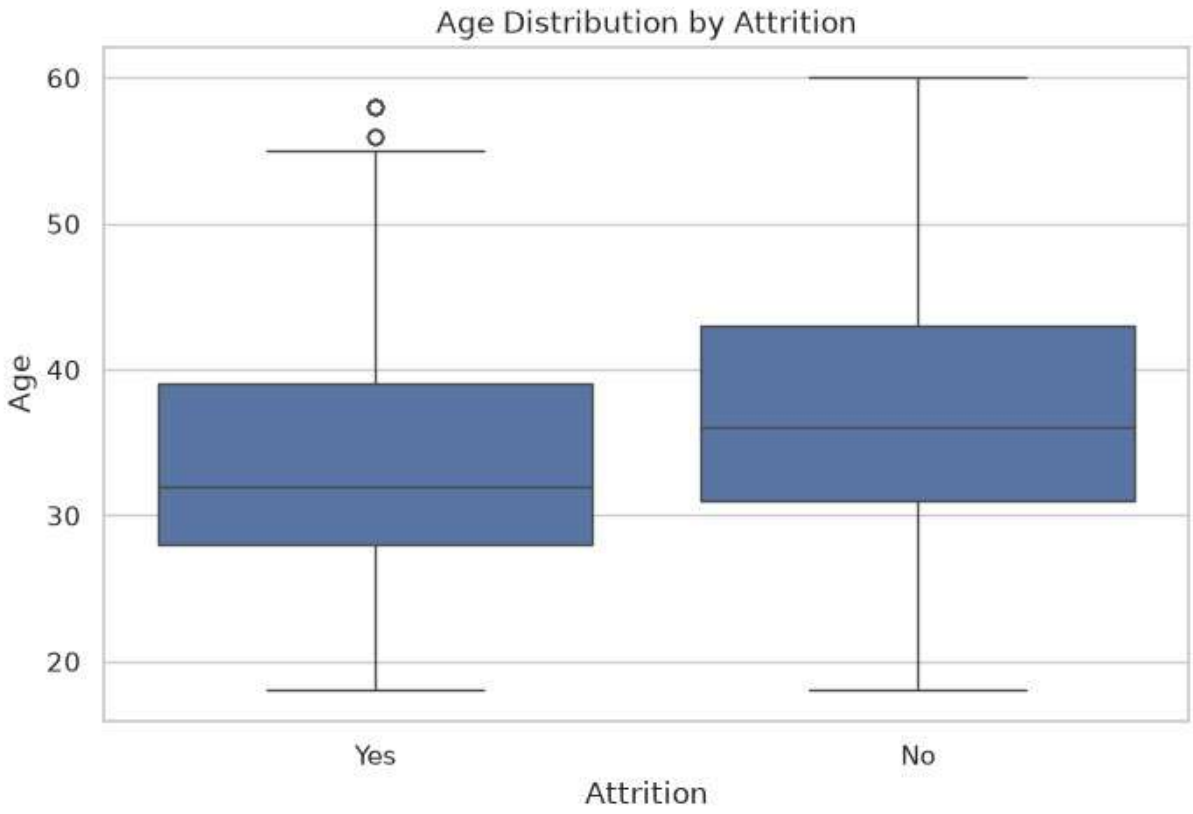
plt.title("Distribution of Employee Age")
plt.xlabel("Age") plt.ylabel("Frequency")
plt.show()
```




```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Attrition", y="Age")

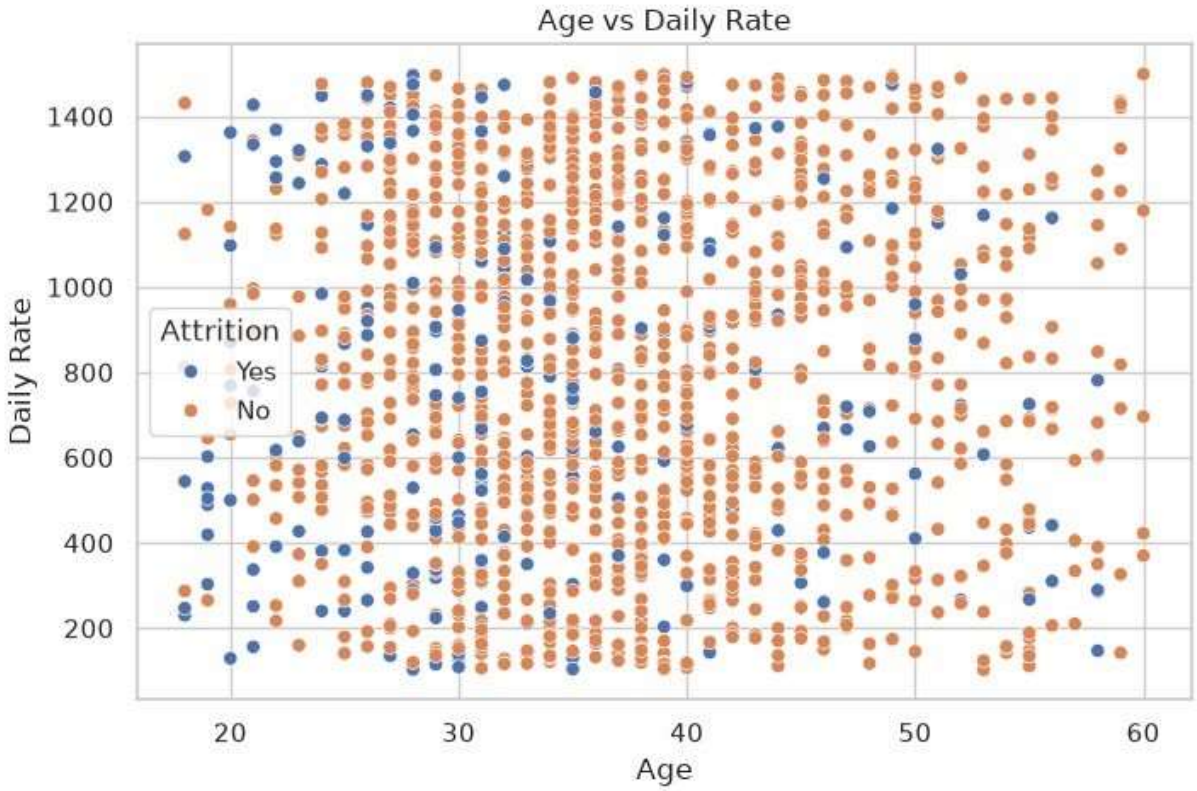
plt.title("Age Distribution by Attrition")
plt.xlabel("Attrition") plt.ylabel("Age")
plt.show()
```




```
plt.figure(figsize=(8, 5))

sns.scatterplot(
data=df,
x="Age",
y="DailyRate",
hue="Attrition"
)

plt.title("Age vs Daily Rate")
plt.xlabel("Age")
plt.ylabel("Daily Rate")
plt.show()
```

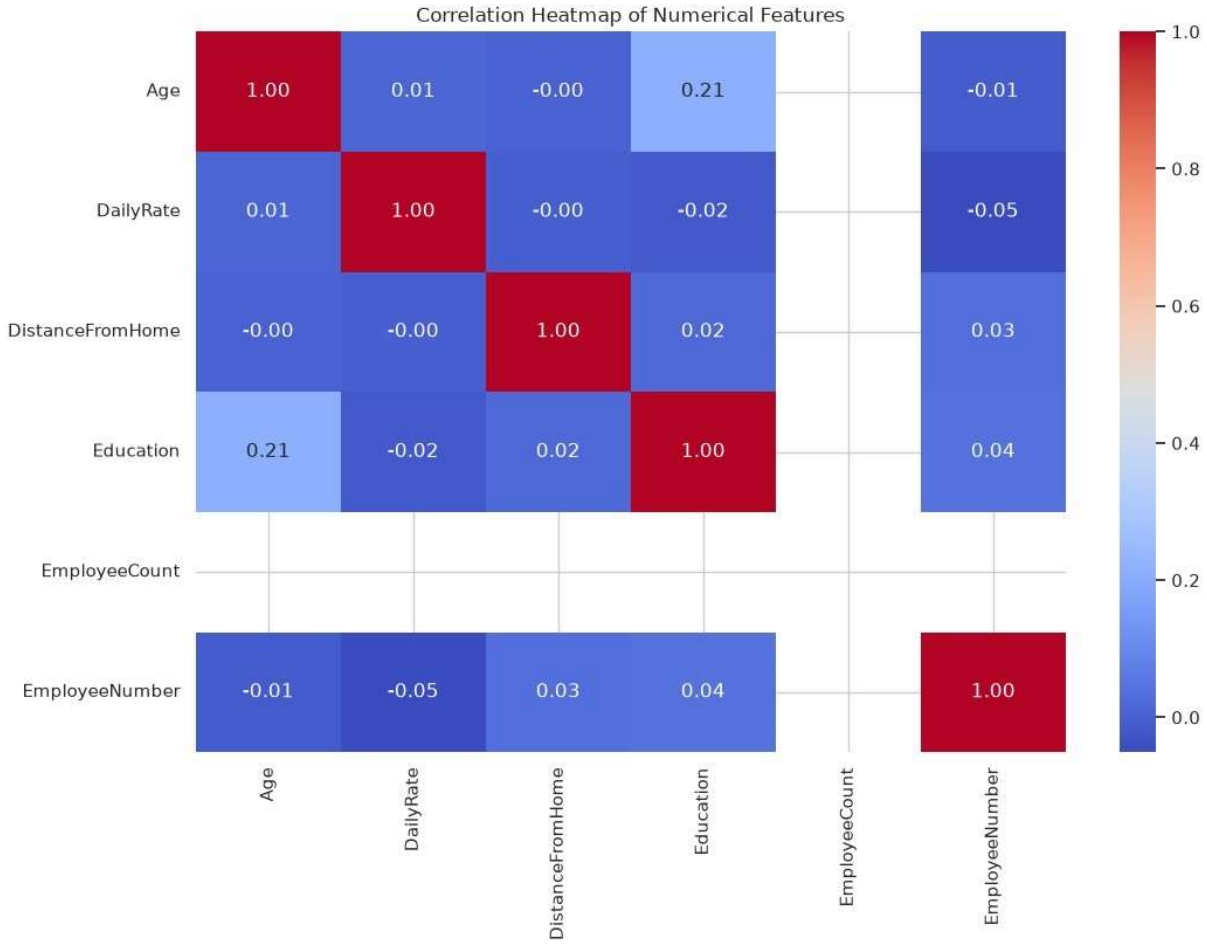



```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(12, 8))

sns.heatmap(
numeric_df.corr(),
annot=True,
cmap="coolwarm",
fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```

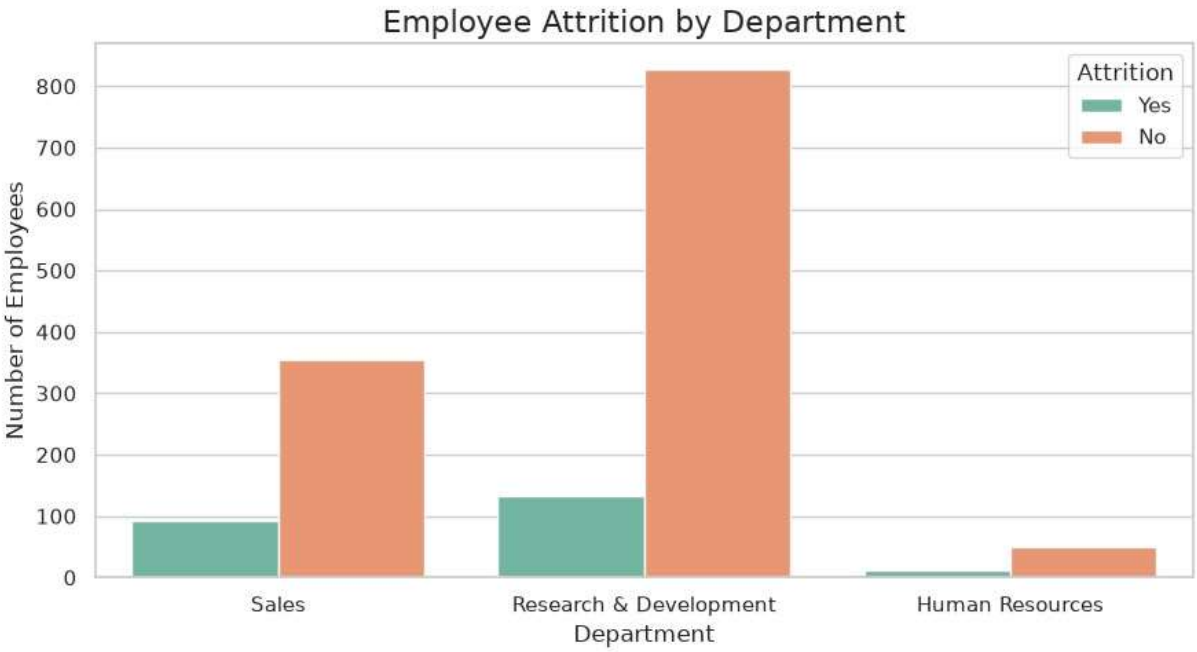



```
plt.figure(figsize=(9, 5))

sns.countplot(
data=df,
x="Department",
hue="Attrition",
palette="Set2"
)

plt.title("Employee Attrition by Department", fontsize=16)
plt.xlabel("Department", fontsize=12) plt.ylabel("Number
of Employees", fontsize=12) plt.legend(title="Attrition")

plt.tight_layout()
plt.show()
```


Employee Attrition by Department
Attrition
Yes
No
Number of Employees
800
700
600
500
400
300
200
100
0
Sales
Research & Development
Human Resources
Department

```python
print("Total Employees:", len(df))

print("\nDepartment-wise Employees:")
print(df["Department"].value_counts())

print("\nAttrition Distribution:")
print(df["Attrition"].value_counts())
print("\nAverage Age:", df["Age"].mean())

print("\nAverage Daily Rate:", df["DailyRate"].mean())
```

```
Total Employees: 1470

Department-wise Employees:
Department
Research & Development    961
Sales                     446
Human Resources            63
Name: count, dtype: int64

Attrition Distribution:
Attrition
No     1233
Yes     237
Name: count, dtype: int64

Average Age: 36.923809523809524

Average Daily Rate: 802.4857142857143
```